Tutorial - 1

A smart Farming system predicts weather, a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters.

Dataset:

Sunlight hours (x_1)	water in liters (x_2)	Bias (b)	Target (y)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

Given bias = b = 1

Initial ~~learning~~ weights - $w_1 = 0$, $w_2 = 0$

η = learning rate = 0.1

$$\begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

Sample (1): $x_1 = 3$, $x_2 = 1$, $b = 1$ $\boxed{y = 0}$

$\boxed{z = \text{weighted sum} = w_1 x_1 + w_2 x_2 + b}$

$z_1 = (0)(3) + (0)(1) + 1$

$= 1$

$\hat{y} = 1$ $\boxed{y \neq \hat{y}}$

So updating weights

$w_1(\text{new}) = w_1(\text{old}) + \eta(y - \hat{y})x_1$

$w_1(\text{new}) = 0 + 0.1(0 - 1)3$

$= 0.1(-3) = -0.3$

$w_2 = 0 + 0.1(-1)(1) = -0.1$

$w_1 = -0.3$

$w_2 = -0.1$

Sample (2): $(8, 4)$ $(w_1, w_2) = (-0.3, -0.1)$ $\boxed{y = 1}$

$z_2 = (-0.3)(8) + (-0.1)(4) + 1$

$= -2.4 - 0.4 + 1$

$= -1.8$

$z < 0 \Rightarrow \hat{y} = 0$

$\boxed{y \neq \hat{y}}$

updating weights:

$w_1 = (-0.3) + (0.1)(1-0)(8) = 0.5$

$w_2 = (-0.1) + (0.1)(1)(4) = 0.3$

Sample 3: $(5, 2)$ $(w_1, w_2) = (0.5, 0.3)$ $\boxed{y=0}$

$z_3 = (0.5)5 + (0.3)2 + 1$

$z_3 = 4.1 \geq 0$

$\hat{y} = 1$ $\therefore \boxed{y \neq \hat{y}}$

updating weights

$w_1 = 0.5 + (0.1)(0-1)5 = 0$

$w_2 = 0.3 + (0.1)(-1)2 = 0.1$

Sample 4: $(10, 5)$ $(w_1, w_2) = (0, 0.1)$ $\boxed{y=1}$

$z_4 = 10(0) + 0.1(5) + 1$

$= 1.5$

$\boxed{\hat{y}=1}$ $\therefore \boxed{y = \hat{y}}$

no need to update weights go with the latest updated weights.

Sample 5: $(7, 3)$ $(w_1, w_2) = (0, 0.1)$ $\boxed{y=1}$

$z_5 = 7(0) + 3(0.1) + 1$

$z_5 = 2.3 \geq 0$

$\boxed{\hat{y}=1}$

$\therefore \boxed{y = \hat{y}}$

Final weights are

$w_1 = 0$

$w_2 = 0.1$